UNIVERSITY OF UYO

FACULTY OF SCIENCE

DEPARTMENT OF MATHEMATICS

SESSION: 2023/2024

FIRST SEMESTER EXAMINATION

TIME: 3HRS

COURSE CODE: MTH311 (MATHEMATICAL METHODS III)

INSTRUCTION: ANSWER ANY FOUR (4) QUESTIONS ONLY

1a. State the conditions that guarantee the existence and uniqueness of solution of the initial value problem

$$\dot{x}(t) = f(t,x,y), \quad \dot{y}(t) = g(t,x,y),, \qquad (1.0)$$
$$x(t_0) = x_0, \quad y(t_0) = y_0$$

b.. Using the Picard's iterative method, show that the solution of the initial value problem $\dot{x}(t) = 2t(1+x)$, $x(0) = 0$, exists and is unique.

2a. Express the following differential equations as a Sturm-Liouville equation; -

i. $x^2\ddot{y} + x\dot{y} + (\lambda x^2 - m^2)y = 0$, ii. $\ddot{y} + \frac{2x}{1-x^2}\dot{y} + \frac{u}{1-x^2}y = 0$,

bi. Solve the Sturm-Liouville problem

$$\ddot{y}(x) + \lambda y(x) = 0, \; 0 < x < L, \; y(0) = y(L) = 0. \qquad (2.1)$$

ii. Show that the solution satisfies the orthogonality property.

3i. Evaluate the series solution of the differential equation $\ddot{y} + y = 0$, $-\infty < x < \infty$ near the ordinary point $x_0 = 0$

ii. Show that solution is unique

4 Define ordinary and singular points of the following equations

I. $(x - \frac{\pi}{2})^2\ddot{y} + (\cos x)\dot{y} + (\sin x)y = 0$, ii. $x^2(1-x)\ddot{y} + (x-2)\dot{y} - 3xy = 0$ (iii) $4x\ddot{y} + 2\dot{y} + y = 0$

b. Evaluate the solution of the differential equation $x^2\ddot{y} + x\dot{y} + y = 0$, near the point $x_0 = 0$.

5a. Evaluate the series solution of the initial value problem $2x^2\ddot{y} - x\dot{y} + (1+x)y = 0$ near the point $x_0 = 0$

6a. Derive the Bessel function $J_{-\nu}(x)$ using the Bessel equation

$$x^2\ddot{y} + x\dot{y} + (x^2 - \nu^2)y = 0, \; \nu \geq 0$$

evaluated about the analytical point $x_0 = 0$

b. Hence, solve the differential equation $x^2\ddot{y} + x\dot{y} + (x^2 - 1)y = 0$, near $x_0 = 0$.

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